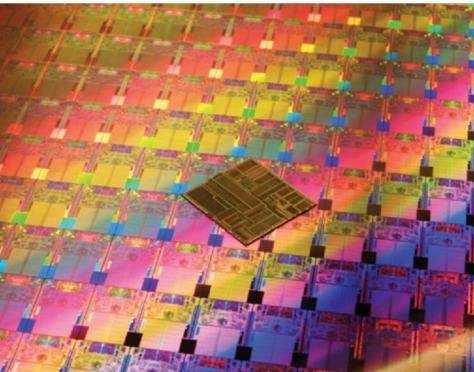


# MULTI-CORE AND OTHER TRENDS IN THE EMBEDDED WORLD

**Multi-core wave is engulfing the mobile phone world too. Here is a quick overview of this as well as other major trends in the embedded space**



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One are the days when embedded systems were a part of complex industrial and defence equipment alone. Today, these have found place in washing machines, microwave ovens, cars and, thanks to mobile phones, in almost everybody's pockets too! Quite naturally, the increase in demand and applications continues to fuel innovations in embedded systems technology on an almost daily basis. This article brings to you some of the significant recent developments in this field.

## 1 Multi-core magic in mobiles and other devices

"One key trend that is taking the world by storm is the rise of 'multi-core.' This year's *Consumer Electronics Show* played host to multiple mobile devices running on dual-core CPUs, and soon we will see devices further enhanced with tri-core and multi-core processors," says Guru Ganesan, managing director, ARM India.

In a recent report titled 'Multi-core Components and Tools,' published by the Venture Development Corporation, it was indicated that 8.4 per cent of the respondents were using a multi-core

architecture on their current project. Of those currently using single-processor architectures, 14.3 per cent reported that they expected to be using multi-core processor architectures within the next two years.

There are many reasons why multi-core processors are slowly replacing single-core processors. Each processor in a multi-core system works independently, allowing the user to multi-task with ease, speed and flexibility. It also works well when the CPU needs to perform complex steps, as the multiple cores work together to complete these tasks faster. Further, multi-core processors are also more power-efficient, so the users are able to do more without having to charge their phones frequently. This also helps the industry meet various energy- and environment-related goals and regulations.

"For these reasons, especially from the power consumption aspect, I believe multi-core processors will be a game-changer in the electronics industry. Dual-core is quickly becoming mainstream and devices running on quad-core processors are likely to see light in 2012. I believe that such technology step-ups on processor designs will increase the capabilities of smartphones. It certainly seems like multi-core is here to stay and is the stepping stone to 'many-core' processors," says Ganesan.

In order to harness the full potential of such processors, we also need real-time operating systems and applications that are designed from the ground-up to think and act as per the parallel processing philosophies. There is no point in treating a multi-core like